ATIE

Safe market routing for Sri Lankan produce

ModelX 2.0 - Two-Page Summary | Team 404 Brain not found | Informatics Institute of Technology

ATIE predicts wholesale crop prices 48-72 hours ahead and turns them into a safe market-routing decision. It compares net profit after fuel, terrain, spoilage, handling, and vehicle costs, so a higher headline price never becomes an unsafe recommendation.

Horizon	Model DS	MAE	SM-FPR	Reroutes
48h	67.89%	6.05	3.00%	6,502
72h	68.74%	6.62	4.96%	6,669

The problem

Smallholder farmers and informal traders often decide where to send produce without knowing whether a local market will be glutted when the consignment arrives. A higher quoted price can still be the wrong choice after transport, spoilage, terrain, handling, and vehicle costs.

The product

The dashboard takes origin, default market, crop, quantity, vehicle, and horizon. It returns one directive, the recommended destination, every market's profit ledger, weather context, and the source and freshness of the data.

User workflow

- Configure the consignment and default market.
- Compare every reachable market by safe net profit.
- Act on a reroute, proceed, or hold directive.

Canonical scenario

1,000 kg of tomatoes from Anuradhapura, defaulting to Dambulla. The fixture is visibly labeled `DEMO_FIXTURE` and reports:

Dambulla forecast	**LKR 220.00 -> 154.00**
Recommended	**Colombo Pettah**
Profit delta	**LKR 23,238.58**
Safe uplift	**15.84%**

The same request with `demo_mode: false` runs normal model inference and returns `PROCEED_DEFAULT` in the current data state.

How it works

```
React dashboard
      |
Spring Boot API ----> PostgreSQL
      |
Python ML service ----> Random Forest
      ^
Ingestion worker ----> HDX / weather / CROPIX
```

The ML service forecasts prices. The backend owns the money and the safety decision:

```
uplift = (NP_alt - NP_default) / NP_default * 100
reroute only when uplift >= 5%
and delta >= LKR 1,000
```

Why it is safe

- Net profit includes effective volume after spoilage, round-trip terrain-adjusted fuel, and handling.
- Decimal arithmetic is used for every routing decision.
- A non-positive default profit never manufactures a positive uplift.
- The frontend never recomputes the decision or uplift.

What ships

- A deployable Compose stack with five health-checked services.
- A documented API, Swagger UI, fallback path, and provenance endpoint.
- A browser-tested dashboard with explicit source and freshness labels.

Reliability and evidence

- Model, fallback snapshot, and canonical fixture are checked into the repository.
- Failed live calls are visibly labeled and never presented as live answers.
- API responses carry `X-Request-ID`; health and metrics are exposed.
- Optional API-key protection is disabled by default for judging.
- CI covers ML tests, backend tests, frontend lint/build, and a Compose-backed Playwright test.

Failure behavior

If the ML service is unavailable, the backend can use a deterministic fallback with a visible `FALLBACK` label. If the live analysis call fails, the dashboard preserves the verified fixture only with an explicit offline warning.

Decision contract

Condition	Directive
Safe candidate passes	REROUTE
Default profit > 0	PROCEED
Default profit <= 0	HOLD

Evaluation

SM-FPR is the rate at which a reroute recommendation loses to the default market after realized holdout prices are substituted. It is a routing backtest, not a price-error proxy.

Baseline 48h	DS	MAE	SM-FPR
Model	67.89%	6.05	3.00%
Naive	0.00%	6.98	4.43%
MA7	46.67%	9.41	7.25%

What is still limited

CROPIX needs a working configured endpoint. HDX is monthly and historical. The model is seed-backed until a supervised retraining pipeline is added. SM-FPR covers a finite, versioned route matrix and does not claim every possible road or consignment.

Safety outcome

A false positive is treated as a catastrophic failure. The system would rather proceed locally or hold a harvest than send a farmer to a distant market that loses after real costs.

Validation

- The canonical fixture and live model path are both exercised end to end.
- The Postgres schema validates against the running Compose database.
- The ML suite covers contracts, fallbacks, features, routing, and evaluation.

Run it

```
cp .env.example .env
docker compose up --build -d --wait
open http://localhost:15173
```

Swagger: `http://localhost:8080/swagger-ui.html` Provenance: `http://localhost:8080/api/v1/metadata/data-status`

Evidence

- Canonical fixture returns `CRITICAL_REROUTE`, Pettah, LKR 23,238.58, and 15.84%.
- Live model mode returns seven market forecasts and a visible `MODEL` badge.
- The browser test verifies the fixture-to-model transition, route ledger, forecast trace, and weather panel.
- All five Compose services reach healthy state.

Data honesty

Every source is labeled `LIVE`, `CACHED`, `SEED`, or `UNAVAILABLE`. The current verified snapshot is `PARTIAL`: Open-Meteo is live and current; WFP/HDX is live but stale, with supported wholesale observations ending April 2021; CROPIX is unavailable.

The model remains trained on checked-in seed history. Live ingestion refreshes weather, supported historical rows, and provenance; it does not silently retrain the model or relabel retail data as wholesale.

Operational checks

Control	Status
Postgres schema	Validated
Frontend proxy	Verified
Request tracing	Enabled
Data provenance	Visible
Security headers	Verified

Repository

The normative rules are in `CONTRACTS.md`; implementation and verification details are in `README.md` and `IMPLEMENTATION_PLAN.md`. The final dashboard capture is in `docs/evidence/live-dashboard.png`.

Demo path

- Load the Northern glut fixture and inspect the safe reroute.
- Run the same consignment without demo mode to show live model output.
- Read the provenance panel and forecast trace before the final decision.

ATIE
Safe routing first.